

Ruiyang (Ryan) Ye

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EDUCATION

University of Waterloo | Cumulative Average: 92.90

Sept. 2025 – Apr. 2030 (Expected)

Bachelor of Mathematics in Combinatorics and Optimization, Pure Math, and Statistics

Waterloo, ON

PROJECTS

SlopFilter | *JavaScript, html, python*

May. 2026

- A web browser extension that actively hides slop content according to user model
- Uses pyTorch to train for video cover and title slop detection, actively inferences video thumbnails with ONNX
- Also removes slop-like clickable containers with css
- Link to Github repo: SlopFilter

IMC Prosperity | *Python Algorithmic Trading*

Apr. 2026

- Finished top 5% as freshman, ranked 22 overall in Canada. My algorithm ranked 14 in Canada.
- Implemented successful trading algorithms with arbitrage, EMA, mean-reversion, and market making

Formalization of Elementary Number Theory | *Abstract Algebra, Number Theory*

Sep. 2025 – Dec.2025

- Implemented an axiomatic integer algebra in Rocq, proving ring properties from minimal axioms
- Proven results include Euclidean division, Bézout's lemma, and Gauss's lemma
- Completed **1582** lines of code, finishing with unique prime factorization in general rings.
- Link to Github repo: Formalization-of-Elementary-Number-Theory

Origametry | *Abstract Algebra, Projective and Combinatorial Geometry, Analysis*

Sep. 2025 – Present

- Studied structures and properties of origami as a formal mathematical system
- Proved around 50 theorems ranging from Abstract Algebra to Analysis
- Combined hobby with rigorous mathematical proofs
- experience in modern mathematical methods of axiomatisation and generalization

Origami Pattern Designer | *Java*

Sep. 2024 – Jan. 2025

- Constructed data structure in symmetry to formal math system
- Implements creases as primitive graph structures
- Inspired by Hatori-Huzita and D.Nedrenco constructions

EXPERIENCE

Contest Math Tutor

Sep. 2024 – Present

Self-employed

Calgary, AB

- Provides group and one-on-one tutoring on contest math
- Material varies from AMC 8 to AIME and Olympiad level
- Around 500 hours of tutoring in total
- Explained problem-solving strategies to improve student understanding and confidence

Peer Mentor

Nov. 2024 – Jun. 2025

Calgary Bridge Foundation for Youths

Calgary, AB

- Enhanced communication skills of mentees, offering workshops on interpersonal communication.
- Contributed to implementing referral system among current students.
- Over 150 hours of participation and teaching.
- Improved mentee engagement with campus resources by compiling and sharing comprehensive guide.

AWARDS

Waterloo Trading Competition

2026

WQC club

University of Waterloo, ON

* Scored 25th as freshman among 112 undergraduates selected from 1700 applicants

Canadian Mathematics Olympiad - Participant

2025

Canadian Math Society

Canada

Alberta High School Mathematics Competition - First Place

2025

University of Alberta

Edmonton, AB

Canadian Senior Math Competition - Honor Roll Group II

2024

Top 21

Canada Lynx Math Competition - Perfect Mark

2025

Top 7

TECHNICAL SKILLS

Languages: Rocq, Lean, Java, Javascript, Python, C, SQL, Haskell

Developer Tools: Git, VS Code, Excel, Microsoft Azure, Google Cloud Platform

Libraries: pyTorch, pandas, NumPy, Matplotlib, ONNX